

Code Documentation

Contemporary hybridization and localized genomic differentiation between grey seal subspecies

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The main dataset used here included whole-genome sequences from 119 grey seal (*Halichoerus grypus*) individuals.

Read data from 17 Arctic ringed seal (*Pusa hispida*) individuals from Greenland were added to serve as an outgroup in the *f*-statistics estimation.

All analyses were performed in the CSC Computing Environment – IT Center for Science, Finland.

Setup

For simplicity, it is assumed that all steps are executed from the current directory and that all tools are available in the bin directory.

```
work=$(pwd)
tools=$work/bin
rmasker=$tools/RepeatMasker
seqbility=$tools/seqbility
bedtools=$tools/bedtools
busco=$tools/busco
miniprot=$tools/miniprot
bwa_mem2=$tools/bwa-mem2
samtools=$tools/samtools
gatk3=$tools/gatk3
gatk4=$tools/gatk4
bcftools=$tools/bcftools
vcftools=$tools/vcftools
beagle=$tools/beagle
genmulti=$tools/generate_multihetsep.py
msmc2=$tools/msmc2
combinecc=$tools/combineCrossCoal.py
msmc_im=$tools/MSMC_IM.py
plink=$tools/plink
smartpca=$tools/eigensoft/bin/smartpca
convertf=$tools/eigensoft/bin/convertf
admixture=$tools/admixture
king=$tools/king
deeptools=$tools/deeptools
minimap2=$tools/minimap2
nextflow=$tools/nextflow
swiftover=$tools/swiftover
blast=$tools/blast
snpeff=$tools/snpEff*
bigWigToBedGraph=$tools/ucsc-tools/bigWigToBedGraph
```

seqbility contains splitfa, gen_raw_mask.pl, gen_mask and makeMappabilityMask.py, blast contains makeblastdb and blastp and deeptools contains bamCoverage.

1. Positive mask and functional categories

1.1 Positive mask

The repetitive elements in the reference genome were identified with RepeatMasker and dog repeat libraries, creating the output \${refname}.fa.out.gff:

```
$rmasker -xsmall -gff --engine rmbblast -species "Canis familiaris" ${refname}.fa
```

The positive mask was created with Heng Li's SNPable, following the instructions of Stephan Schiffels. The mappable regions were identified by splitting the reference genome into 75 bp fragments and realigning them to the reference genome. Based on that, a FASTA file with mappable regions was created with gen_raw_mask.pl and gen_mask, and a BED file was created with makeMappabilityMask.py.

```

$seqbility/splitfa ${refname}.fa 75 | split -l 20000000

for i in $(ls x??); do bwa aln -R 1000000 -O 3 -E 3 ${refname}.fa $i > $i.sai; done

for i in $(ls x??); do bwa samse ${refname}.fa $i.sai $i > $i.sam; done

cat x??sam | $seqbility/gen_raw_maks.pl > rawMask.fa

$seqbility/gen_mask -l 35 -r 0.5 rawMask.fa > mask.fa

python makeMappabilityMask.py

for i in $(ls posmask); do zcat posmask/$i | bgzip -c >> posmask.bed.gz; done

```

The repeat elements (negative mask) and the mappable regions (positive mask) were combined to a joint mask:

```
$bedtools subtract -a posmask.bed -b ${refname}.fa.out.gff > joint_mask.bed
```

1.2 Functional categories

BUSCO analysis

The BUSCO analysis was performed using the lineage “carnivora”.

```
$busco -i ${refname}.fa -m genome -l carnivora
```

Gene models in the ribbon seal reference were inferred with miniprot using the elephant seal protein sequences.

```
$miniprot --gff ${refname}.fa elephant_seal_protein.faa > elephant_seal_genes_tmp.gff
```

BUSCO generated an additional set of gene models, which were extracted from the BUSCO output.

```
cp BUSCO_${refname}.fa/logs/miniprot_align_carnivora_odb12_out.log > busco_genes_tmp.gff
```

GFF files were sorted and filtered by keeping only gene models with 80%+ identity.

```

grep -v "##" ${gffset}_tmp.gff | sort -k1,1 -k4,5n > ${gffset}.gff

grep mRNA ${gffset}.gff | cut -f9 | sed 's/[;=]/ /g' | awk '$6>=0.8{print $2}' > ${gffset}_keep.txt

grep -f ${gffset}_keep.txt ${gffset}.gff > ${gffset}_keep.gff

```

Matches to BUSCO CDS were identified and extracted.

```

grep CDS elephant_seal_genes_keep.gff | gff2bed | cut -f-3 | bedtools sort -i - | uniq \
> elephant_seal_keep_CDS.bed

grep CDS busco_genes_keep.gff | gff2bed | cut -f-3 | bedtools sort -i - > busco_genes_keep_CDS.bed

```

```
$bedtools intersect -a <(grep CDS elephant_seal_genes_keep.gff) -b busco_genes_keep_CDS.bed \
-f 0.9 -F 0.9 | gff2bed | bedtools merge -i - -c 10 -o first > first.bed

cut -f4 first.bed | sed 's/[;=]/ /g' | cut -d" " -f2 | uniq | sort | uniq > keepIDs.txt

grep -f keepIDs.txt elephant_seal_genes_keep.gff > elephant_seal_BUSCO_genes.gff
```

To obtain corresponding Ensembl gene IDs and gene symbols for downstream analyses, the elephant seal proteins were aligned against ferret protein sequences (MusPutFur1.0.pep.all; Ensembl version 115) using BLAST.

Build a database of ferret protein sequences

```
$blast/makeblastdb -in Mustela_putorius_furo.MusPutFur1.0.pep.all.faa -dbtype prot \
-parse_seqids -out ferret_protein
```

Blast and keep first match with at least 75% identity.

```
$blast/blastp -db ferret_protein -query elephant_seal_protein.faa \
-outfmt "6 qseqid sseqid pident length evalue bitscore stitle" \
-evalue 1e-7 -num_threads 8 -out blast_out.tsv

less blast_out.tsv | awk '{if($3>75 && done[$1]!=1){print $0;done[$1]=1}}' \
> blast_out_75.tsv
```

2. WGS data processing

2.1 Mapping and variant calling

These steps use BWA-MEM2 for mapping the data in FASTQ format to the reference genome, SAMtools for processing the output files and GATK for realignment around indels and variant calling.

```
$bwa_mem2 mem -R${readgroup} -K 100000000 -Y ${refname}.fa ${fastq1}${fastq2}
| $samtools view -h - | $samtools fixmate -m - ${sample_id}_mapped.bam

$samtools sort ${sample_id}_mapped.bam -Obam -o ${sample_id}_aligned.bam

$gatk3 -T RealignerTargetCreator \
-R ${refname}.fa \
-I ${sample_id}_aligned.bam \
-o targets.intervals

$gatk3 -T IndelRealigner \
-R ${refname}.fa \
-I ${sample_id}_aligned.bam \
-targetIntervals targets.intervals \
-o ${sample_id}_realigned.bam

$samtools markdup ${sample_id}_realigned.bam ${sample_id}_markdup.bam
```

```
$gatk4 HaplotypeCaller \
-R ${refname}.fa \
-I ${sample_id}_markdup.bam \
-O ${sample_id}.gvcf.gz \
-ERC GVCF
```

2.2 Joint calling

These steps use GATK.

A map file listing GVCFs for *all* samples was created

```
for gvcf in *.gvcf.gz; do
sample_id=$(basename "$gvcf" .gvcf.gz)
echo -e "$sample_id\t$(realpath "$gvcf")"
done > samples.map
```

Single-sample GVCFs were imported into GenomicsDB and jointly genotyped

In practice, joint genotyping was done in batches for computational efficiency, by breaking up the genome in smaller intervals and specifying `-L $batch` for each.

```
$gatk4 GenomicsDBImport \
--genomicsdb-workspace-path database \
--sample-name-map samples.map

$gatk4 GenotypeGVCFs \
-R ${refname}.fa \
-V gendb://database \
-O grey_full.vcf.gz
```

In addition, the combined Greenland ringed seal GVCFs were jointly genotyped, including only sites present in the filtered grey seal VCF

Note: the filtering process is described below.

```
$gatk4 GenotypeGVCFs \
-R ${refname}.fa \
-V ringed.gvcf.gz \
-L grey_final.vcf.gz \
-O ringed.vcf.gz \
--force-output-intervals
```

And merged with the grey seal VCF

```
$bcftools merge grey_final.vcf.gz ringed.vcf.gz -Oz -o merged_grey_ringed.vcf.gz
```

2.3 Filtering

Sites fixed for the same ALT allele across samples were removed

The resulting VCF was used in the SnpEff analysis.

```
$bcftools view -q 0.0001:minor grey_full.vcf.gz -Oz -o grey_seg.vcf.gz
```

Only regions included in the positive mask were retained

For this, the positive mask described above was used.

```
$bcftools view -T joint_mask.bed grey_seg.vcf.gz -Oz -o grey_seg_positive.vcf.gz
```

Biallelic SNPs were kept

```
$bcftools view -v snps -m2 -M2 grey_seg_positive.vcf.gz \
-Oz -o grey_seg_positive_snp.vcf.gz
```

Filtering based on sequencing depth

The maximum and minimum values were set based on the the total mapping coverage (across all samples) distribution.

```
$bcftools view -i 'INFO/DP>1000 && INFO/DP<3500' \
-Oz -o grey_final.vcf.gz grey_seg_positive_snp.vcf.gz
```

A version of the VCF with stricter filtering was also created to use in the Identity-by-state (IBS) distance estimation.

```
$bcftools view -i 'INFO/DP>1400 && INFO/DP<3500' \
-Oz -o grey_final_strict.vcf.gz grey_seg_positive_snp.vcf.gz
```

2.4 Sequencing depth estimation in 50bp windows

Sequencing depth was estimated along the largest contig in 50bp windows.

```
cat sample_dir.txt | while read smp dir; do
    $samtools depth -a -r Hfa001 $dir/${smp}_markdup.bam \
    | awk '{s+=3;if($2%50==0){OFS="\t";print $1,$2,s/50;s=0}}' > $smp.tsv;
done
```

3. Demographic analyses with MSMC2

Grey seal data were phased with beagle, separately for Baltic and Atlantic grey seals.

```
dir=msmc/ne_phasing

$beagle nthreads=5 ne=100000 gl=${file}.vcf.gz out=${dir}/${file}_imputed.vcf.gz
$beagle nthreads=5 ne=100000 gt=${dir}/${file}_imputed.vcf.gz out=${dir}/${file}_phased.vcf
```

The two phased VCF files were merged into `grey_phased.vcf.gz` with `bcftools`. Then, the combined data were converted to multihetsep.

```

for chr in {001..499}; do
  CHR=Hfa$chr
  len=`grep -w $CHR $FAI | cut -f2`
  if [ $len -gt 100000 ]; then
    list=""
    for SMP in $SAMPLES; do
      $bcftools view -s $SMP -r $CHR $VCF -Oz -o ctgs/$SMP.$chr.vcf.gz
      list=$list" "msmc/ctgs/$SMP.$chr.vcf.gz
    done
    $genmulti --mask mask/$CHR.mask.bed.gz $list > multi/chr${chr}.multihetsep.txt \
    2> multi/log.txt
    rm $list
  fi
done

```

For the MSMC2 analyses, 6 samples of each subspecies were used.

For the demographic inference, MSMC2 within populations was run with four individuals per analysis. e.g., one run for the Atlantic:

```

$msmc2 -p 27*1+1*2+1*3 -I 0,1,2,3,4,5,6,7 -o output4_atl/ABCD multi/chr*.multihetsep.txt \
&> output4_atl/ABCD.runlog

```

MSMC2 within populations was also run for two individuals per analysis, to combine with the across-population runs.

```

$msmc2 -p 27*1+1*2+1*3 -I 0,1,2,3 -o output2_atl/AB multi/chr*.multihetsep.txt &> \
output2_atl/AB.runlog
$msmc2 -p 27*1+1*2+1*3 -I 12,13,14,15 -o output2_bal/GH multi/chr*.multihetsep.txt &> \
output2_bal/GH.runlog

```

MSMC2 across populations was run with two individuals from each population. e.g.:

```

$msmc2 -p 27*1+1*2+1*3 \
-I 12-0,12-1,12-2,12-3,13-0,13-1,13-2,13-3,14-0,14-1,14-2,14-3,15-0,15-1,15-2,15-3 \
-o output4_bal_atl/GAHB multi/chr*.multihetsep.txt &> output4_bal_atl/GAHB.runlog

```

MSMC2 rCCR analysis output files were combined with the corresponding two-sample MSMC2 analyses with combineCrossCoal.

```

$combinecc output4_bal_atl/GAHB.final.txt output2_bal/GH.final.txt \
output2_atl/AB.final.txt > output4_bal_atl/BG_GAHB.final.txt

```

MSMC_IM were then run on the output:

```

$msmc_im -beta 1e-8,1e-6 -mu 1.826e-8 -o output4_ccr/BG_GAHB output4_ccr/BG_GAHB.final.txt

```

4. Genetic variation and population structure

Create input data for PCA and admixture analysis

For the principal component analysis (PCA) and the admixture analysis a thinned version of the grey seal VCF was used. Here, thinning was done so that the distance between the SNPs is at least 1kb.

```
$plink --vcf grey_final.vcf.gz --allow-extra-chr --bp-space 1000 \
--recode vcf bgz --out grey_final_thin
```

The thinned VCF data were converted to BED format.

```
$plink --vcf grey_final_thin.vcf.gz --allow-extra-chr --allow-no-sex \
--make-bed --out grey_final_thin
```

The FAM and BIM files were slightly modified.

```
#Rename the first column of the FAM file. The smp_pop.tsv used here contains a
#sample ID and a corresponding population column (Baltic/Atlantic).

mv grey_final_thin.fam grey_final_thin.fam_

awk 'NR==FNR {pop[$1]=$2; next} ($1 in pop) {$1=pop[$1]} {print}' smp_pop.tsv \
grey_final_thin.fam_ > grey_final_thin.fam

#Make the first column of the BIM file numeric

mv grey_final_thin.bim grey_final_thin.bim_

awk 'BEGIN{OFS="\t"} {ctg = substr($1,4,6) + 0;print ctg,$2,$3,$4,$5,$6}' \
grey_final_thin.bim_ > grey_final_thin.bim
```

4.1 PCA

Data were converted to Eigenstrat format and a PCA was performed.

```
base=grey_final_thin

cat > convert.txt << EOF
genotypename:    $base.bed
snpname:         $base.bim
indivname:       $base.fam
outputformat:    EIGENSTRAT
genotypeoutname: $base.geno
snpoutname:      $base.snp
indivoutname:    $base.ind
EOF

$convertf -p convert.txt
```



```

mv grey_final_thin.ind grey_final_thin.ind_
sed 's/:// /' grey_final_thin.ind_ | awk '{print $2,$3,$1}' > grey_final_thin.ind

cat > smartpca.txt << EOF
genotypename:    $base.geno
snpname:         $base.snp
indivname:       $base.ind
evecoutname:     $base.pca.evec.txt
evaloutname:     $base.pca.eval.txt
usenorm:         NO
EOF

$smartpca -p smartpca.txt > smartpca.out

```

4.2 IBS distance estimation

The IBS distance estimation was done with PLINK and used the VCF file with stricter depth filtering.

```

$plink --vcf grey_final_strict.vcf.gz --distance 1-ibs square gz \
--allow-extra-chr --out grey_final_strict

```

4.3 Admixture analysis

This analysis was done with ADMIXTURE, here for $K=2-5$.

```

base=grey_final_thin

for K in {2..5}; do
$admixture --cv $base.bed $K | tee $base.${K}.log
done

```

4.4 Relatedness estimation

Relationship inference was done with KING. Prior to the estimation, `grey_final.vcf.gz` was converted into BED format, similarly to before. Family IDs were the same for all individuals.

```

$king -b grey_final.bed --related --prefix grey_rel

```

5. Assessment of hybrid status of individual HG301 with f_4 -statistic and f_4 -ratio test

f -statistics were computed with the R package ADMIXTOOLS2. Slightly different setups were used to test for gene flow towards other individuals/populations.

Prior to this, the merged grey-ringed VCF was converted to BED format with PLINK and the files were modified as in 4.

```

library(tidyverse)
library(admixtools)

# Define individuals and populations

codes_smp <- read.table("merged_grey_ringed.fam")

codes_smp <- codes_smp[, c(1,2)]

colnames(codes_smp) <- c("code", "smp")

## Keep HG301 as its own "population", but leave other DEN_ROD as "DEN_ROD"

codes_smp[codes_smp$code == "DEN_ROD" & codes_smp$smp == "HG301", "code"] <- "DEN_ROD_301"

## Pool populations

codes_smp[startsWith(codes_smp$code, "ICE"), "code"] <- "ICE"

codes_smp[startsWith(codes_smp$code, "UK"), "code"] <- "UK"

## SWE_STA + FIN + EST (Baltic populations that did not undergo extinction; OLD_BAL)

codes_smp[startsWith(codes_smp$code, "FIN") | startsWith(codes_smp$code, "EST"), "code"] <- "OLD_BAL"

codes_smp[codes_smp$code == "SWE_STA", "code"] <- "OLD_BAL"

extract_f2(pref = "merged_grey_ringed", outdir = "f2_grey_ringed",
            inds = codes_smp$smp, pops= codes_smp$code)

# Compute f4 statistics

test_ind <- "DEN_ROD_301"

## From the Atlantic

f4.res_atl <- f4(f2_blocks, "OLD_BAL", test_pop_ind[1], "UK","GRE_RIN")

## From the Baltic

f4.res_bal <- f4(f2_blocks, "UK", test_pop_ind[1], "OLD_BAL","GRE_RIN")

# Perform f4-ratio test

## alpha for Atlantic origin. Population order: A, O, X, C, B

f4r.atl <- qp4ratio(f2_blocks,as.matrix(tibble("ICE","GRE_RIN",test_ind, "OLD_BAL", "UK")))

```

```
## alpha for Baltic origin. Population order: A, O, X, C, B

f4r.bal <- qpf4ratio(f2_blocks,as.matrix(tibble("OLD_BAL","GRE_RIN",test_ind, "UK", "SWE_FAL")))
```

6. Sex linked contigs and sex identification

For detecting the X contigs, three complementary approaches were used: coverage ratio comparison between males and females (m:f), heterozygosity comparison and homology to the dog X chromosome and the Saimaa ringed seal X-linked scaffolds.

6.1 Coverage comparison

This approach follows Catalán et al. (2025).

Coverage was estimated in 10-kbp windows with deepTools for 5 samples of each morphologically determined sex.

```
cat selected_sample_dir.txt | while read smp dir; do
    $deeptools/bamCoverage -b $dir/${smp}_markdup.bam -bs 10000 --normalizeUsing RPKM --verbose \
    --outFileFormat bedgraph -o ${smp}_10kb_RPKM.bedgraph
done
```

The bedgraphs were combined into a single BED file using `bedtools unionbedg`.

Then, computations were then done in R.

```
library(dplyr)
library(readr)
library(tidyr)

cov <- read.table("selected_samples_10kb_RPKM.bed", header=TRUE)

# Sex info for the selected individuals

sexinfo <- read_tsv("smp.sex.tsv", col_names = c("ind", "sex"))

# After determining the the five largest contigs showing no notable coverage differences
# between samples, we used their average coverage (across windows of the five contigs)
# per individual to normalize the coverage of each window for that individual.

biggest_auto_chrs <- c("Hfa001", "Hfa002", "Hfa003", "Hfa004", "Hfa005")

auto_mean <- cov %>%
  filter(chrom %in% biggest_auto_chrs) %>%
  summarise(across(-c(chrom, start, end), mean, na.rm = TRUE))

cov_norm <- cov
for (s in colnames(cov_norm)[-(1:3)]) {
  cov_norm[[s]] <- cov_norm[[s]] / as.numeric(auto_mean[[s]])
}
```

```

}

# Add sex info and perform a non-parametric Wilcoxon Rank Sum Test to test for significant
# differences in contig coverage values between sexes, applying a Bonferroni multiple
# test correction, as in Catalán et al. (2025).

## Add preliminary sex info to the full cov file

cov_norm_sex <- cov_norm %>%
  pivot_longer(-c(chrom, start, end),
               names_to = "ind",
               values_to = "coverage") %>%
  left_join(sexinfo, by = "ind")

## Perform test

results <- cov_norm_sex %>%
  group_by(chrom) %>%
  summarise(
    p = wilcox.test(coverage ~ sex)$p.value,
    male_mean = mean(coverage[sex == "m"], na.rm = TRUE),
    female_mean = mean(coverage[sex == "f"], na.rm = TRUE)
  ) %>%
  mutate(p_adj = p.adjust(p, method = "bonferroni"))

results <- results %>%
  mutate(mf_ratio = male_mean / female_mean)

# From the contigs that show significant differences, the ones with a m:f
# coverage ratio of 0.5±0.1 were selected

res_x_contigs <- results %>%
  filter(p_adj < 0.05 & mf_ratio >= 0.4 & mf_ratio <= 0.6)

# To verify the assigned sex, the coverage values of single individuals (not only
# the mean of m and f) were also inspected.

```

6.2 Sex identification

One of the candidate X contigs identified with this approach, Hfa209, was observed to contain *TSPYL2*, which in dogs and humans is located on the X chromosome. Males and females in the full dataset were separated by comparing mean coverage values (averaged across windows) for all individuals on this contig.

For this, coverage was estimated in 10-kbp windows for contig Hfa209 in all individuals, as well as for the autosomal contigs Hfa001-Hfa005 (for normalization). Coverage estimates were generated separately for each contig and were then combined into a single file (per individual).

```
regions="Hfa001 Hfa002 Hfa003 Hfa004 Hfa005 Hfa209"

cat sample_dir.txt | while read smp dir; do
  out=${smp}_10kb_RPKM.bedgraph
  > $out
  for r in $regions; do
    bamCoverage -b ${dir}/${smp}_markdup.bam -bs 10000 --normalizeUsing RPKM \
      --region $r --outFileFormat bedgraph -o tmp_${smp}_${r}.bedgraph

    grep -v "^track" tmp_${smp}_${r}.bedgraph >> $out #Remove header and append
    rm tmp_${smp}_${r}.bedgraph # Remove temporary file
  done
done
```

Again, the bedgraphs were combined into a single BED file using `bedtools unionbedg`.

Sex identification was done in R.

```
cov_total <- read.table("all_samples_10kb_RPKM.bed", header=TRUE)

# Normalize, as before

cov_norm_total <- cov_total
for (s in colnames(cov_norm_total)[-1:3]) {
  cov_norm_total[[s]] <- cov_norm_total[[s]] / as.numeric(auto_mean[[s]])
}

# Take average across Hfa209 windows

cov_norm_mean_Hfa209 <- cov_norm_total %>%
  filter(chrom== "Hfa209") %>%
  summarise(across(-c(start, end), mean, na.rm = TRUE))

cov_norm_mean_Hfa209_long <- cov_norm_mean_Hfa209 %>%
  pivot_longer(-chrom, names_to = "ind", values_to = "coverage")

# This can be plotted to separate males and females.
```

6.3 Heterozygosity comparison

First, genetic diversity (π) per individual per site was estimated with VCFtools.

```
for smp in $(cut -f1 metadata.tsv); do
  $bcftools view -s $smp grey_final.vcf.gz | $vcftools --vcf - --site-pi --out $smp
done
```

Heterozygosity (H) in 250-kbp windows was computed in R following Olkkonen and Löytynoja (2023) and contigs with difference in H between the previously identified males and females were detected.

`window_function.R` used here is provided separately.

```

source("window_function.R") # Estimates coverage in 250-kbp windows,
# correcting the window width with the positive mask and nan-sites.

library(ggplot2)
library(dplyr)

pi_est_files <- list.files(pattern = "^HG.*\\.sites\\.pi$")

pi_est_windows_list <- list()

for (i in seq_along(pi_est_files)) {
  filename <- pi_est_files[i]
  data <- read.table(filename, sep="\t", header=TRUE)
  pi_est_windows_list[[filename]] <- windows(data)
  print(paste("Processed", filename))
}

all_windows <- bind_rows(pi_est_windows_list, .id = "ind")

all_windows$ind <- gsub("\\.sites\\.pi$", "", all_windows$ind)

# Heterozygosity in windows for each individual was normalized by the mean H of the
# five largest autosomal contigs.

auto_mean <- all_windows %>% #contains windowed H estimates for all individuals
  filter(ctg %in% biggest_auto_chrs) %>%
  group_by(ind) %>%
  summarise(mean_pi_ind = mean(pi, na.rm = TRUE), .groups = "drop")

all_windows_norm <- all_windows %>%
  left_join(auto_mean, by = "ind") %>%
  mutate(pi_norm = pi / mean_pi_ind)

# Add sex info for all individuals

sexinfo_all <- read_tsv("smp_all.sex.tsv", col_names = c("ind", "sex"))

all_windows_norm_sex <- all_windows_norm %>%
  left_join(sexinfo_all, by = "ind")

# Perform Wilcoxon Rank Sum Test, as before

results_H <- all_windows_norm_sex %>%
  group_by(ctg) %>%
  summarise(
    p = wilcox.test(pi_norm ~ sex)$p.value)

```

```

    male_mean = mean(pi_norm[sex == "m"], na.rm = TRUE),
    female_mean = mean(pi_norm[sex == "f"], na.rm = TRUE)
  ) %>%
  mutate(p_adj = p.adjust(p, method = "bonferroni"))

res_x_contigs_H <- results_H %>%
  filter(p_adj < 0.05)

# The contigs with significant differences between sexes and near zero H in males
# were compared to the X contigs determined by the coverage approach.

```

6.4 Alignment approach

As a third line of evidence, the ribbon seal reference genome was mapped against a dog (ROS_Cfam_1.0.dna_sm.toplevel; Ensembl version 115) and a Saimaa ringed seal genome, using minimap2.

For example, for the Saimaa ringed seal:

```

minimap2 saimaa.fa ${refname}.fa -o ribbon_vs_saimaa.paf

# Add PID column (BLAST-like alignment identity)

awk '{pid=($10/$11)*100; print$0, pid}' ribbon_vs_saimaa.paf > ribbon_vs_saimaa_pid.paf

# Focus on regions mapping to the Saimaa putative X scaffolds

awk '$6~/^scaffold10/' ribbon_vs_saimaa_pid.paf > ribbon_vs_saimaa_X.paf

# Inspect based on PID and alignment (number of matching bases) length
# (e.g., pid>60%, length>10kbp)

column -t ribbon_vs_saimaa_X.paf | awk '$10>10000 && $19>60 {print}' \
> ribbon_vs_saimaa_X_10kbp_pid60.paf

```

Based on the three lines of evidence, a final set of X contigs was compiled.

Calculate the total length of X contigs

```

while read contig; do
  grep -w $contig ${refname}.fa.fai | cut -f1,2
done < X_contigs_final.tsv > X_contigs_final_length.tsv

less X_contigs_final_length.tsv | awk '{sum += $2} END {print sum}'

```

7. Hybrid's parental populations

To identify the hybrid's maternal populations, a PCA using only X-linked contigs and only male samples was performed.

Input data were generated and the output FAM and BIM files were modified as in 4.

```
$plink --vcf grey_final_thin.vcf.gz --allow-extra-chr --allow-no-sex --make-bed \
--chr $x1 $x2 ... \
--remove females.txt \
--out males_x
```

The BIM file was further modified so that the contigs artificially start from number 1.

```
mv males_x.bim males_x.bim_

less males_x.bim_ | awk 'BEGIN{ctg=0;OFS="\t"}{if($1!=pc){ctg+=1}pc=$1;$1=ctg;print $0}' \
> males_x.bim
```

Next, data were converted to Eigenstrat format and a PCA was performed, as in 4.1.

8. Allele frequency estimation and gene functions

8.1 Allele frequency differences (ΔAF)

First, allele frequencies per site were estimated separately for the Baltic and the Atlantic individuals, using VCFtools.

The hybrid (HG301) was excluded from the Baltic samples and the migrant (HG517) from the Atlantic ones.

```
vcftools --gzvcf grey_final.vcf.gz --keep Baltic_no_HG301.tsv --freq --out Baltic
vcftools --gzvcf grey_final.vcf.gz --keep Atlantic_no_HG517.tsv --freq --out Atlantic
```

Then, absolute allele frequency differences were computed in R.

```
# Read data (e.g., for the Baltic)

Baltic_freq <- read.table("Baltic.freq", row.names = NULL)

colnames(Baltic_freq) <- c("CHROM", "POS", "N_ALLELES", "N_CHRB", "AF1B", "AF2B")
#AF1 corresponds to the REF and AF2 to the ALT allele.

# Make freq numeric

Baltic_freq$AF1B <- as.numeric(sub(".*:", "", Baltic_freq$AF1B))
Baltic_freq$AF2B <- as.numeric(sub(".*:", "", Baltic_freq$AF2B))

# Merge with the data for the Atlantic and compute absolute ΔAF

Bal_Atl_freq <- merge(Baltic_freq[, -c(3,4)], Atlantic_freq[, -c(3,4)],
                      by=c("CHROM", "POS"), sort=FALSE)

Bal_Atl_freq$AF_diff <- abs(Bal_Atl_freq$AF1B - Bal_Atl_freq$AF1A)

write.table(Bal_Atl_freq, "Bal_Atl_AF_diff.tsv", quote=FALSE, col.names=TRUE,
            row.names = FALSE, sep="\t")
```


Visualization in chromosomes

To visualize ΔAF in chromosomes, the values were lifted over to a dog genome.

For this, a liftover chain between the ribbon seal and the dog genome was generated using the Nextflow Liftover pipeline. The pipeline is automatically fetched from GitHub by `nextflow`, while helper tools had been installed locally.

```
#Get dog genome

wget https://ftp.ensembl.org/pub/release-115/fasta/canis_lupus_familiaris/dna/Canis_lupus_familiaris.ROS_Cfam_1.0.dna_sm.toplevel.fa.gz

#Create seal-to-dog chain file

gunzip Canis_lupus_familiaris.ROS_Cfam_1.0.dna_sm.toplevel.fa.gz

dog=Canis_lupus_familiaris.ROS_Cfam_1.0.dna_sm.toplevel

$nextflow run evotools/nf-LO --source ${refname}.fa --target ${dog}.fa --distance far \
--aligner minimap2 --outdir seal_to_dog --no_netsynt
```

Next, the ΔAF values were transferred to the corresponding coordinates of the dog genome using `swiftover` (<https://github.com/blachlylab/swiftover>).

```
# Convert the ΔAF TSV file into a BED file

less Bal_Atl_AF_diff.tsv | awk -F"\t" 'BEGIN {OFS="\t"} \
{print $1, $2-1, $2, $3"&"$4"&"$5"&"$6"&"$7}' > Bal_Atl_AF_diff.bed

# Transfer values

$swiftover -t bed -c $seal_to_dog_chain -i Bal_Atl_AF_diff.bed \
-u unmatched_seal_to_dog.bed > dog_AF_diff.bed

mv dog_AF_diff.bed dog_AF_diff.bed_

less dog_AF_diff.bed_ | sed 's/&/\t/g' > dog_AF_diff.bed
```

8.2 Genes and functions

To inspect genes in proximity to the areas of high differentiation, we used the functional region annotation described above.

Genes within 1 Mbp of regions of high allele frequency difference ($\Delta AF \geq 0.55$) were extracted and then filtered based on Gene Ontology (GO) terms, retaining those with annotations related ion and osmotic regulation or lipid metabolism. For this, GO annotations for ferret gene IDs (MusPutFur1.0) were retrieved from Ensembl using the R package `biomaRt`.

```
library(biomaRt)
library(dplyr)
library(tidyr)
```

```

# Select database and dataset

ensembl <- useEnsembl(
  biomart = "genes",
  dataset = "mpfuro_gene_ensembl"
)

# Ensembl gene IDs to get GO terms for

genes_t <- read.table("genes_AF_diff_55_1Mbp.tsv", quote = "", sep = "\t",
  col.names=c("contig", "start", "end", "protein_ncbi", "description",
    "ensembl_protein", "ensembl_gene_id_version", "gene_symbol"))

genes <- genes_t[,7] # Get column with Ensembl IDs

# Build query and get GO IDs and names

go_list <- getBM(attributes=c("ensembl_gene_id_version", "go_id", "name_1006"),
  filters = "ensembl_gene_id_version",
  values = genes,
  mart=ensembl)

# Remove rows with no GO names and get all terms per gene ID

go_list_clean <- go_list %>%
  mutate(name_1006 = na_if(name_1006, ""))

go_list_clean <- go_list_clean %>%
  filter(!is.na(name_1006))

go_per_gene <- go_list_clean %>%
  group_by(ensembl_gene_id_version) %>%
  summarise(
    go_id = paste(unique(go_id), collapse = ";"),
    go_name = paste(unique(name_1006), collapse = ";"))

# Merge with gene info

genes_with_go <- genes_t %>%
  left_join(go_per_gene, by = "ensembl_gene_id_version")

#Filter for specific names

pattern <- paste(c("lipo", "lipid", "sterol", "osmo", "sodium", "potassium", "chloride",
  "\\banion\\b", "\\bcation\\b", "ion transport", "ion channel",

```

```

"monoatomic ion"), collapse = "|")

go_filtered <- genes_with_go %>%
  filter(grepl(pattern, go_name, ignore.case = TRUE))

```

9. Variant effect prediction and conservation scores

Before predicting the functional effect of variants segregating between the subspecies, the ancestral state of alleles was inferred and the VCF was repolarized. The version of the VCF repolarized was not masked and included both biallelic SNPs and biallelic indels.

The ancestral state inference and repolarization of the VCF were done using custom scripts.

Script to add AA information:

```

cat > scripts/addAA.sh << 'EOF'

vcf=$1
temp=$2

bam=mirang_sorted.bam
ref=Ribbon_PR_18_6_2024.fa

$gatk4 HaplotypeCaller -I $bam -R $ref -L $vcf.vcf.gz -O temp/$temp.gvcf.gz \
--output-mode EMIT_ALL_CONFIDENT_SITES -ERC BP_RESOLUTION

$gatk4 GenotypeGVCFs -R $ref -L $vcf.vcf.gz --call-genotypes true \
--force-output-intervals $vcf.vcf.gz -V temp/$temp.gvcf.gz -O temp/$temp.vcf.gz

$bcftools query -i 'INFO/DP>4' -f '%CHROM\t%POS\t%REF\t%ALT[\t%GT]\n' temp/$temp.vcf.gz \
| awk '{OFS="\t";printf "%s.%09d\t", $1,$2; print $0}' > temp/$temp.1.sites.tsv
$bcftools query -f '%CHROM\t%POS\t%REF\t%ALT\n' $vcf.vcf.gz \
| awk '{OFS="\t";printf "%s.%09d\t", $1,$2; print $0}' > temp/$temp.2.sites.tsv

join -j1 temp/$temp.2.sites.tsv temp/$temp.1.sites.tsv \
| awk '$4==$8 && ($5==$9 || $9=="."){
    OFS="\t";
    if($10=="0/0" || $10=="0|0") {
        print $2,$3,$4,$5,$4
    } else if($10=="1/1" || $10=="1|1") {
        print $2,$3,$4,$5,$5
    } else {
        print $2,$3,$4,$5, "."
    }
}' > temp/$temp.AA.sites.tsv

bgzip temp/$temp.AA.sites.tsv
tabix -s1 -b2 -e2 temp/$temp.AA.sites.tsv.gz

echo '##INFO=<ID=AA,Number=1,Type=Character,Description="Ancestral allele">' \
> temp/$temp.hdr.txt

```

```

$bcftools annotate -a temp/$temp.AA.sites.tsv.gz -c CHROM,POS,REF,ALT,INFO/AA -h \
temp/$temp.hdr.txt -Oz -o ${vcf}_AA.vcf.gz $vcf.vcf.gz

$bcftools view -m2 -M2 ${vcf}_AA.vcf.gz | perl scripts/polarizeVCF.pl \
| bcftools view - -Oz -o ${vcf}_bial_pol.vcf.gz
$bcftools index ${vcf}_bial_pol.vcf.gz
EOF

```

Script to polarize VCF data:

```

cat > scripts/polarizeVCF.pl << 'EOF'

while(<>) {
    if(/^#/) {
        print $_;
        next;
    }
    chomp;
    my($chr,$site,$rs,$ref,$alt,$info) = (split(/\t/, $_))[0,1,2,3,4,7];
    my $ancestral_allele;
    $ancestral_allele = (split(/AA=/, $info))[1];
    $ancestral_allele = (split(/\|/, $ancestral_allele))[0];
    $ancestral_allele = uc($ancestral_allele);

    if($ancestral_allele eq $ref) { #ok as it is
        print "$_\n";
        next;
    } else { #Switch alleles
        my @tmp = split(/;/, $info);
        foreach my $t ( @tmp ) {
            next if($t !~ /AF/);
            my($key,$val) = split(/=/, $t);
            my $new_AF = 1-$val;
            $_ =~ s/$key=$val/$key=$new_AF/;
        }
        my($chr,$site,$rs,$ref,$alt,$qual,$filter,$info,$format,@tmp) = split(/\t/, $_);
        my $swp = $ref;
        $ref = $alt;
        $alt = $swp;
        my $str = join "\t", $chr, $site, $rs, $ref, $alt, $qual, $filter, $info, $format, "";
        foreach my $t ( @tmp ) {
            if($t =~ /[01][\|\\|][01]/) {
                my $sep='/';
                if($t =~ /\|/) { $sep='|'; }
                #($t) =~ s/\|/\\/;
                my($a,$b,$c,$d,$e) = (split(/\|:./, $t));
                if($a == 0) { $a = 1; }
                elsif($a == 1) { $a = 0; }
                if($b == 0) { $b = 1; }
                elsif($b == 1) { $b = 0; }

                if($a == 1 && $b == 0) { $a = 0; $b = 1; }
            }
        }
    }
}
EOF

```

```

        $t = "$a$sep$b:$d,$c:$e";
    }
    $str .= "$t\t";
}
$str =~ s/\t$//;
print "$str\n";
}
}
EOF

```

These scripts were utilized to add AA and re-polarize sites where ‘INFO/AA=ALT’.

```
bash scripts/addAA.sh grey_seg.vcf.gz mirang_seals
```

As multiallelic sites could not be polarized, the polarized *biallelic* sites were subsequently combined with the non-polarized multiallelic sites.

```

$bcftools isec -p isec -c all -Oz grey_seg_bial_pol.vcf.gz grey_seg.vcf.gz

$bcftools concat -a isec/0002.vcf.gz isec/0001.vcf.gz -Ou | $bcftools sort --temp-dir \
tmp_merge -Oz -o full_with_pol.vcf.gz

$bcftools index full_with_pol.vcf.gz

```

Prediction of variant effects was done with SnpEff. To this end, the program files were copied into the local working directory and the input data were structured according to the program’s requirements.

```

mkdir -p SnpEff/data/ribbonseal

cp $snpeff SnpEff

cp ${refname}.fa genes.gff SnpEff/data/ribbonseal

cd SnpEff/data/ribbonseal

mv ${refname}.fa sequences.fa

```

Next, the config file was edited and an annotation database was created.

```

echo "ribbonseal.genome : ribbonseal" >> snpEff.config

java -jar snpEff.jar build -gff3 -v ribbonseal -noCheckCds -noCheckProtein

```

And the VCF file was annotated.

```

file_ann=grey_full_with_pol_ann

( java -jar snpEff.jar -v -stats $vcf/${file_ann}.html ribbonseal $vcf/full_with_pol.vcf.gz \
| bcftools view -Oz -o $vcf/${file_ann}.vcf.gz ) &> $vcf/${file_ann}.log

```

The VCF was additionally annotated with **Genomic Evolutionary Rate Profiling (GERP)** scores obtained from Ensembl datasets for ferret.

First, the GERP scores were retrieved and converted to BED format.

```
wget https://ftp.ensembl.org/pub/release-115/compara/conservation_scores/\
92_mammals.gerp_conservation_score/gerp_conservation_scores.mustela_putorius_furo.MusPutFur1.0.bw

$bigWigToBedGraph gerp_conservation_scores.mustela_putorius_furo.MusPutFur1.0.bw \
ferret_conservation_scores.bed
```

Then, a ferret-to-ribbon seal liftover chain was created and the GERP scores were lifted over to the ribbon seal genome, similarly to 8.1.

```
# Get ferret genome and create chain file

wget https://ftp.ensembl.org/pub/release-115/fasta/mustela_putorius_furo/dna/\
Mustela_putorius_furo.MusPutFur1.0.dna_sm.toplevel.fa.gz

gunzip Mustela_putorius_furo.MusPutFur1.0.dna_sm.toplevel.fa.gz

ferret=Mustela_putorius_furo.MusPutFur1.0.dna_sm.toplevel

$nextflow run evotools/nf-L0 --source ${ferret}.fa --target ${refname}.fa --distance far \
--aligner minimap2 --outdir ferret_to_seal --no_netsynt

# Transfer scores

$swiftover -t bed -c $ferret_to_seal_chain -i ferret_conservation_scores.bed \
-u unmatched_ferret_to_seal.bed > ribbon_conservation_scores.bed
```

With this, the VCF could be annotated.

```
echo '##INFO=<ID=GERP,Number=1,Type=Float,Description="GERP conservation score">' \
> gerp_header.txt

$bcftools annotate -a ribbon_conservation_scores.bed -h gerp_header.txt \
-c CHROM,FROM,TO,INFO/GERP -Oz -o grey_full_with_pol_ann_GERP.vcf.gz \
grey_full_with_pol_ann.vcf.gz
```

References

Catalán, A., Gygax, D., Candolin, U., Tusso, S., Duchon, P., Höhna, S., 2025. Sex-biased Migration and Demographic History of the Big European Firefly *Lampyrus noctiluca*. *Mol. Biol. Evol.* 42, msaf123. <https://doi.org/10.1093/molbev/msaf123>

Olkkonen, E., Löytynoja, A., 2023. Analysis of population structure and genetic diversity in low variance Saimaa ringed seals using low-coverage whole-genome sequence data. *STAR Protoc.* 4, 102567. <https://doi.org/10.1016/j.xpro.2023.102567>